Elliott Scott M.S.

Data Scientist | Machine Learning Engineer
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Portfolio: https://elliottscott774.github.io/

Summary

Data Scientist with 2+ years of experience designing, building, and deploying end-to-end machine learning systems. Expert in developing robust ETL pipelines, implementing advanced deep learning models, and creating interactive analytics platforms. Proven ability to translate complex data from diverse sources into actionable intelligence and data-driven products.

Experience

Graduate Data Analyst — IU Indianapolis • *2024 – 2025*

- Architected and deployed a multi-page analytics platform using Streamlit, Docker, and AWS S3, enabling self-service data exploration for non-technical users.
- Conducted a comprehensive analysis of 300+ technology companies by integrating CRM, behavioral, and demographic data to identify key predictors for successful partnership agreements.
- Produced an interactive win-probability Shiny application based on live data streams, and engineered advanced performance metrics that influenced future data-collection protocols.

Data Science Consultant — Booz Allen Hamilton • *2021 – 2023*

- Developed a Python-based ETL pipeline for high-frequency networking logs.
- Responded to daily ad-hoc data requests, creating visual dashboards that highlighted key performance metrics.
- Presented monthly insights to leadership, translating complex model outputs into clear, non-technical narratives.

Research Intern — U.S. Army NBC Battle Lab • *Summers 2018 & 2019*

- Built an electric-drone battery prediction system using PCA and linear regression; improved endurance forecasts by 11 %.
- Designed and executed comprehensive performance testing protocols for multi-vendor radio systems, collaborating with team lead to evaluate functionality, reliability, and usability; produced detailed analysis and recommendation report to inform purchasing decisions.

Skills

- **Languages:** Python, R, SQL
- **Data Science:** pandas, NumPy, scikit-learn, TensorFlow/Keras, PyTorch, statsmodels, NetworkX
- **Visualization:** Plotly, Streamlit, Shiny, Tableau, matplotlib, seaborn, ggplot2
- **Cloud & DevOps:** Docker, AWS (EC2, S3, Lambda), Git/GitHub, CI/CD pipelines, Backblaze B2
- **Databases:** MySQL, Microsoft Access
- **Domain Experience:** NLP, Time-Series Analysis, Recommendation Systems, ETL, Web Scraping

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Projects

- **Deep Learning Author Classifier:** Architected an end-to-end author classification system, training a bidirectional LSTM model in PyTorch on the Project Gutenberg corpus and deploying it as an interactive Streamlit application to identify an author from a text sample.
- **Performance Analytics Platform:** Built a multi-page, Dockerized web application for performance analysis using Streamlit and Plotly. The platform features user authentication and integrates with AWS S3 for data ingestion. Engineered an ETL pipeline to process and merge data from numerous raw sources, creating over 50 normalized metrics and KPIs for analysis and visualization.
- **Asset Valuation & Recommendation System:** Developed a recommendation engine to identify undervalued assets and simulate their impact on a portfolio. The system uses a Decision Tree model trained on web-scraped data to predict performance. Deployed the model as an interactive Streamlit dashboard that allows users to test different asset acquisitions and see the projected impact, with data hosted on Backblaze B2.
- **Network Analysis of Key Connectors:** Quantified the contributions of key connectors within a network using network and spatial analyses on a large event-based dataset. The project introduces a novel composite metric to measure efficiency, combining spatial dominance, volume, and forward progression to identify critical entities that bridge different clusters within the network.
- **Sequence Prediction of Object Trajectories using Seq2Seq LSTM:** Created a Seq2Seq LSTM model using TensorFlow/Keras to predict route trajectories from time-series tracking data (2M+ rows).

Education

- **M.S. Applied Data Science, Sports Analytics Specialization** — *Indiana University Indianapolis* — May 2025
  Relevant coursework: Sports Analytics, Machine Learning, Deep Learning, Statistical Learning, Cloud Engineering & Architecture, Database Design, Data Visualization
- **B.S. Computer Science** — *Towson University* — December 2020

Certifications & Interests

- **Machine Learning** — Stanford Online
- **Top Secret Security Clearance** *(inactive)*
- Former Head Coach — Susquehanna Football Club
- Avid sailor — completed a full circumnavigation of Bermuda.